HCLTech

Name	**Seela Srinivas Chakravarthi**
Designation	Senior Software Engineer – GenAI & Agentic AI
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Professional Profile

Brief Profile	● Senior Software Engineer specializing in Generative AI, Agentic AI, and Full Stack Development, with hands-on experience building enterprise AI solutions using LangChain, LangGraph, LLMs, and vector databases. Skilled in developing and evaluating LLM applications, RAG pipelines, and multi-agent workflows using frameworks such as RAGAS and DeepEval. Knowledgeable in evaluation metrics including faithfulness, answer relevancy, context precision, context recall, task completion, and tool-call accuracy, with an understanding of human-in-the-loop validation and Phoenix observability for assessing response quality and agent performance. ● Experienced in building scalable AI-powered applications using Python, React.js, Node.js, MongoDB, and cloud technologies, with expertise in prompt engineering, semantic search, and AI workflow orchestration. Passionate about developing

	production-ready AI solutions, optimizing business processes through intelligent automation, and continuously learning emerging technologies in Artificial Intelligence and Generative AI. ● Strong foundation in software engineering, system design, REST APIs, and backend development, with the ability to deliver robust, scalable, and user-centric applications aligned with business objectives.
Total Experience	1 Year (GenAI & Agentic AI)
Education	B.Tech CSE (AI & ML - Data Science), Lovely Professional University (2021–2025)
Certifications	GenAI trainings – AWS cloud service, Amazon Bedrock, AWS AgentCore, S3, EC2, AWS console
Core Technology & Skill Matrix	**Generative AI & LLMs** • Large Language Models: Llama 3.2, Qwen, Gemini • Prompt Engineering and Embeddings • Retrieval-Augmented Generation (RAG) Architectures • Stateful AI Agents and Multi-Agent Workflows • MCP Servers and AI Tool Integration **LLM, RAG & Agentic AI Evaluation** • Evaluation Frameworks: RAGAS, DeepEval • LLM Metrics: Answer Relevancy, Correctness, Hallucination, Groundtruth Dataset Creation • RAG Metrics: Faithfulness, Context Precision, Context Recall, Rubrics metrices and scoring • Agent Metrics: Task Completion, Tool-Call Accuracy, Agent testing, Latency of workflows. • Human-in-the-Loop Review and Validation • Observability and Tracing: LangSmith, Phoenix • Agent Latency Monitoring and Error Analysis **AI Frameworks, SDKs & Cloud** • LangChain, LangGraph

	• AWS Bedrock and Amazon Bedrock AgentCore Runtime • Sarvam AI APIs: STT, TTS, Translation, Chat • AWS: Lambda, Cognito, DynamoDB, S3, IAM, API Gateway • Vector Databases: ChromaDB **Full Stack Development** • Frontend: React.js, JavaScript • Backend: FastAPI, Node.js, Express.js • Programming: Python • Databases: SQL, MongoDB

Sample Project – POC

(1).

Project Name	**AI Business Intelligence – BizInsight**
Skill	Generative AI, Stateful Multi-Agent Systems, LangChain, LangGraph, MCP, RAG, SQL, FastAPI, React, Gemini, Phoenix Observability, LLM/RAG/Agentic AI Evaluation, RAGAS, DeepEval, Evaluation Metrics, and Human-in-the-Loop Validation.
Project Description: Built a conversational business intelligence assistant using SQL data, RAG, and stateful multi-agent workflows for query routing, retrieval, and response generation. Integrated MCP database tools, a React interface, and Phoenix tracing to monitor agent execution, latency, and errors. Incorporated LLM, RAG, and	

agent evaluation using RAGAS and DeepEval, covering faithfulness, answer relevancy, context precision, context recall, task completion, and tool-call accuracy, with human-in-the-loop review to validate response quality.

Key Learnings:

- Designing and orchestrating multi-agent workflows using LangGraph.
- Maintaining conversation history with stateful agents.
- Integrating MCP tools to access SQL business data.
- Building RAG pipelines using PDFs and ChromaDB.
- Generating SQL queries from natural language questions.
- Developing FastAPI REST APIs and connecting them to React.
- Visualizing agent execution and tool usage in the UI.
- Tracking agent latency and errors using Phoenix traces.
- Understanding retrieval and response evaluation metrics using RAGAS and DeepEval.

(2).

Project Name	**AI-Powered Unified Multilingual Citizen Service Assistant**
Skill	Python, FastAPI, LangChain, LangGraph, RAG, Agentic AI, Sarvam APIs, Vector Databases, FAISS, Speech-to-Text, Text-to-Speech, Translation APIs, PostgreSQL, REST APIs, AWS Knowledge Base, AWS S3, AWS EC2.

Project Description:

Designed and implemented an AI-powered multilingual Citizen Service Assistant leveraging Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI frameworks to provide citizens with a unified digital interface for accessing government services. The solution enables users to interact through both text and voice channels while automatically detecting their preferred language

Key Learnings:

- Gained hands-on experience in designing, developing, and deploying AI Agents and Generative AI applications for real-world citizen service use cases.
- Implemented Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) to provide accurate, context-aware, and reliable responses from multiple knowledge sources.
- Developed expertise in agentic AI frameworks for workflow automation, task orchestration, and intelligent decision-making.
- Learned prompt engineering techniques to improve response quality, reduce hallucinations, and enhance user interactions.

- Integrated AI solutions with enterprise data sources, APIs, and backend workflows to enable seamless information retrieval and service delivery.
- Built multilingual conversational AI capabilities with automatic language detection, enabling inclusive text and voice-based user experiences.
- Implemented Speech-to-Text (STT) and Text-to-Speech (TTS) functionalities to support voice-enabled interactions.

(3).

Project Name	**AI-Powered Beneficiary Fraud Detection System using RAG and Generative AI**
Skill	Python, FastAPI, React.js, JavaScript, REST APIs, Google Gemini 2.5 Flash, Retrieval-Augmented Generation (RAG), ChromaDB, Sentence Transformers,OCR (Optical Character Recognition)

Project Description:

Developed an AI-powered Government Beneficiary Verification System using **Generative AI, LLMs, RAG, and AI agents** to automate citizen application verification. Integrated **OCR and Gemini 2.5 Flash** to extract and normalize data from Aadhaar cards, income certificates, and application forms, enabling cross-document validation and fraud detection. Implemented a **RAG pipeline using ChromaDB and Sentence Transformers** to retrieve government scheme guidelines and assess applicant eligibility. Built automated verification workflows, risk scoring, and explainable recommendations for officers, along with a dashboard for case review, feedback management, and verification monitoring. Continuously optimized prompts, retrieval accuracy, and model performance to improve decision-making efficiency.

Key Learnings:

- Gained hands-on experience in building and deploying GenAI-powered applications for real-world government service workflows.
- Developed expertise in implementing LLM-based document understanding and intelligent information extraction from unstructured documents.
- Learned advanced OCR integration techniques for processing and validating data across multiple document types.
- Implemented Retrieval-Augmented Generation (RAG) architectures using ChromaDB and Sentence Transformers for policy and guideline retrieval.
- Developed knowledge of vector embeddings, semantic search, and knowledge retrieval optimization.
- Improved understanding of human-in-the-loop workflows by incorporating officer feedback and review processes into the verification lifecycle.

(4).

Project Name	**FinGenie – AI-Powered Voice Expense Tracker**
Skill	React.js, Node.js, MongoDB, Gemini API, NLP, Recharts, Speech Recognition, Node.js, Express.js

Project Description:

Developed **FinGenie**, an AI-powered personal finance assistant using **Generative AI, LLMs, and intelligent automation** to simplify expense tracking through natural language interactions. Built conversational workflows that allow users to record expenses via voice or text, automatically extracting transaction details such as amount, category, merchant, and date. Designed AI-driven expense categorization, spending analytics, and personalized financial insights to help users monitor and optimize their spending habits. Integrated speech-to-text processing, backend data services, and automated pipelines to enable real-time expense tracking and dashboard updates. Continuously optimized prompts, extraction accuracy, and model performance to improve user experience and financial recommendations.

- **Key Learnings:**
- Gained hands-on experience in developing AI-powered conversational applications for personal finance management.
- Implemented Natural Language Processing (NLP) techniques to understand and process voice and text-based expense commands.
- Learned to integrate AI chatbots and LLMs for generating personalized financial insights and user recommendations.
- Developed expertise in speech-to-text processing and voice-enabled user interactions.
- Built automated workflows for expense creation, deletion, tracking, and querying through natural language inputs.
- Designed and implemented interactive dashboards using Recharts for expenditure visualization and trend analysis.
- Gained experience in creating scalable REST APIs for managing financial transactions and user data.
- Learned backend integration techniques for seamless communication between AI services, databases, and frontend applications.

Certifications	GenAI trainings – AWS cloud service, Amazon Bedrock, AWS AgentCore, S3, EC2, AWS console

Declaration

The information presented in this profile is a fair representation of my skills, technical competencies, project experience, leadership responsibilities, and professional achievements. Supporting details can be provided upon request.

Seela Srinivas Chakravarthi
Senior Software Engineer